Math 118: Mathematical Methods of Data Theory

Lecture 20: Dynamic Programming 3

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Let's Play a Game

- We'll flip a coin. Heads = I give you 1 dollar. Tails = you give me one dollar.
- In long run, expect as many heads as tails.
- Suppose I am a cheat!
  - I have two coins, one fair (F) and one loaded (L) .
  - Loaded coin has higher probability of tails:

$$\mathbb{P}[H|F] = \mathbb{P}[T|F] = 0.5 \quad \text{but } \mathbb{P}[H|L] = 0.4 \text{ and } \mathbb{P}[T|L] = 0.6$$

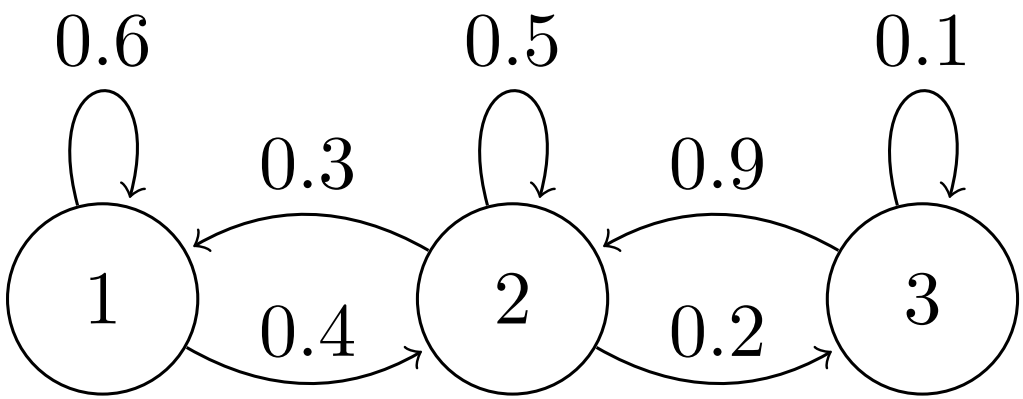
  - I am sneaky: able to switch the coins without you noticing.
- **Question:** How do you predict which coin is currently in use?

Markov Models

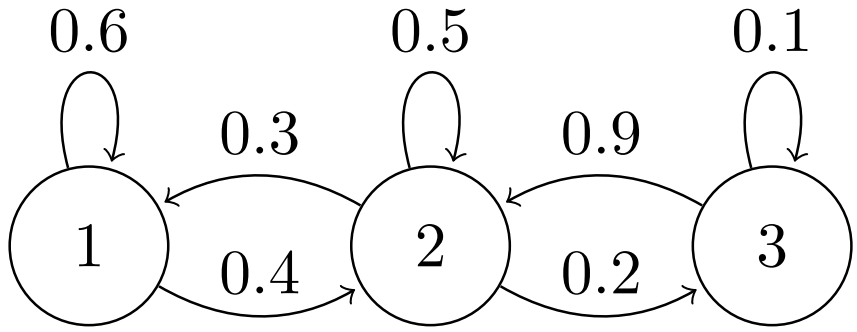
- A Markov Model is (essentially) a weighted, directed graph.
  - Vertices represent **states**.
  - Directed edges represent allowable **state transitions**.
  - Edge weights represent **probability of transition**.
  - **Markov Property:** state x_{n+1} **only** depends on x_n, and not on any previous states.

$$\mathbb{P}\left[x_{n+1} = i | x_1 x_2 \dots x_n\right] = \mathbb{P}\left[x_{n+1} = i | x_n\right]$$

  - Weights of all out edges from any state sum to 1.
  - Self-loops allowed.
- A Markov chain/ process is the sequence of states visited by a random walk on this graph: $x_1 x_2, \dots, x_n \dots$ *(here* $x_i = 1, 2$ *or* 3*)*.



Markov Models



- Initial distribution should be specified, *e.g.*

$$\mathbb{P}[x_1 = 1] = 0.4 \quad \mathbb{P}[x_1 = 2] = 0.3 \quad \mathbb{P}[x_1 = 3] = 0.3$$

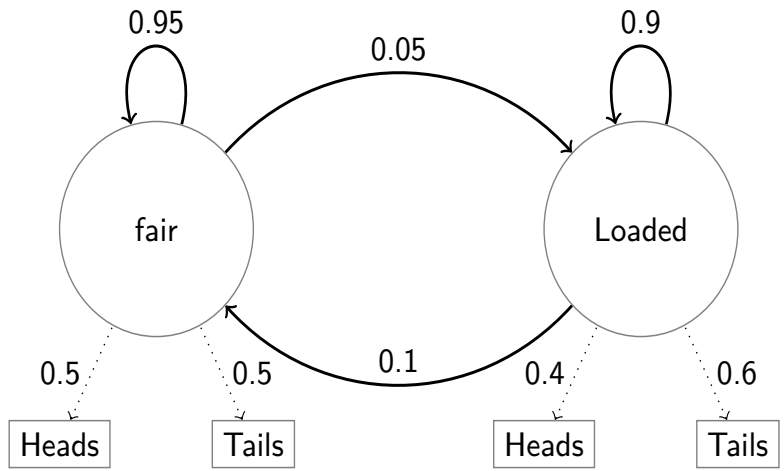
- **Question:** What is probability of observing sequence 12322?

$$\begin{aligned}\mathbb{P}[12322] &= \mathbb{P}[x_5 = 2|1232]\,\mathbb{P}[1232] = \mathbb{P}[x_5 = 2|x_4 = 2]\,\mathbb{P}[1232] = (0.5)\mathbb{P}[1232] \\ &= (0.5)\mathbb{P}[x_4 = 2|123]\,\mathbb{P}[123] = (0.5)\mathbb{P}[x_4 = 2|x_3 = 3]\,\mathbb{P}[123] \\ &= (0.5)(0.9)\mathbb{P}[123] = \ldots \\ &= (0.5)(0.9)(0.2)(0.4)\mathbb{P}[x_1 = 1] = (0.5)(0.9)(0.2)(0.4)(0.4) = 0.0144\end{aligned}$$

- **Question:** Why is this probability so small?
- **Answer:** There are $\sim 3^5$ possible five-step chains. Their probabilities must sum to 1.

Hidden Markov Models

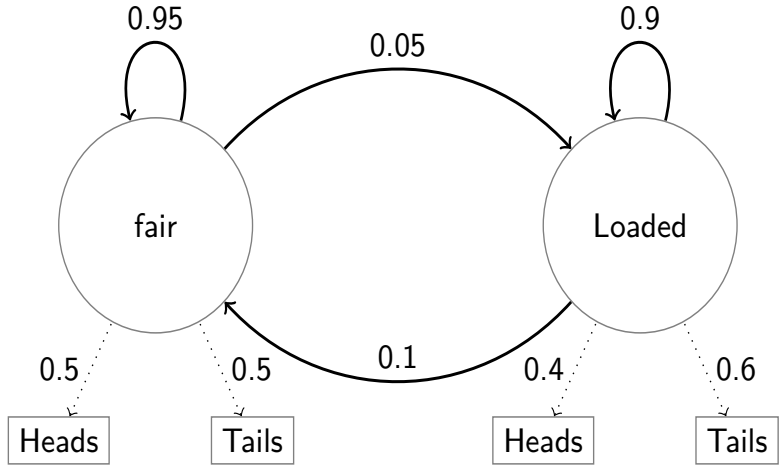
- A hidden MM is one where the current state is hidden from observer.
- Instead, we define a set of **rewards** or **emissions**: $\{e_1, \ldots, e_m\}$
- Each state emits these rewards with different probabilities.
- If at time t model is in state i, observe reward y_j with probability $\mathbb{P}[y_t = e_j | x_t = i]$. Probability of emission **only depends on** x_t.



- **Markov Property:** state x_{n+1} **only** depends on x_n, and not on any previous states **or emissions**.

$$\mathbb{P}[x_{n+1} = i | x_1 \ldots x_n, \pi_1, \ldots \pi_n] = \mathbb{P}[x_{n+1} = i | x_n]$$

Hidden Markov Models



- Initial distribution should be specified, *e.g.*

$$\mathbb{P}[x_1 = F] = 1 \quad \mathbb{P}[x_1 = L] = 0$$

- **Question:** What is probability of states FFL and emissions HHH?

$$\begin{aligned}
\mathbb{P}\left[HHH, FFL\right] &= \mathbb{P}\left[y_3 = H, x_3 = L | HH, FF\right] \mathbb{P}\left[HH, FF\right] \\
&= \mathbb{P}[y_3 = H | x_3 = L] \mathbb{P}[x_3 = L | x_2 = F] \mathbb{P}[HH, FF] \\
&= (0.4)(0.05) \mathbb{P}[HH, FF] \\
&= (0.4)(0.05) \mathbb{P}[y_2 = H | x_2 = F] \mathbb{P}[x_2 = F | x_1 = F] \mathbb{P}[H, F] \\
&= \ldots = (0.4)(0.05)(0.5)(0.95)(0.5) \mathbb{P}[x_1 = F] = 0.00475
\end{aligned}$$

Viterbi's algorithm (for 2 state HMM)

- **Observe:** sequence of emissions $\boldsymbol{Y}_n = y_1 \dots y_n$.
- **Goal:** Find sequence of states $\boldsymbol{X}_n = x_1 \dots x_n$ maximizing the **joint probability** $\mathbb{P}[\boldsymbol{Y}_n, \boldsymbol{X}_n]$.
- We assume Model is **completely known**[1]. That is, we know:
  1. Number of states.
  2. Transition probabilities.
  3. Emission probabilities.
- Define $f(i,k) = \max_{\boldsymbol{X}_{k-1}=x_1 \dots x_{k-1}} \mathbb{P}[\boldsymbol{Y}_k, \boldsymbol{X}_{k-1} | x_k = i]$
- Final answer will be $\max_{\boldsymbol{X}_n} \mathbb{P}[\boldsymbol{Y}_n, \boldsymbol{X}_n] = \max_i f(i,k)$
- **Some Notation:** $A_{ij} = \mathbb{P}[x_{t+1} = i | x_t = j] = \mathbb{P}[i|j]$

[1] If not, we have to **learn** the parameters. This is possible using the Baum-Welch algorithm for example

- **Recall:** $f(i,k) = \max_{\boldsymbol{x}_{k-1}=x_1\ldots x_{k-1}} \mathbb{P}[\boldsymbol{Y}_k, \boldsymbol{X}_{k-1}|x_k = i]$
- For any fixed $\boldsymbol{X}_{k-1}$:

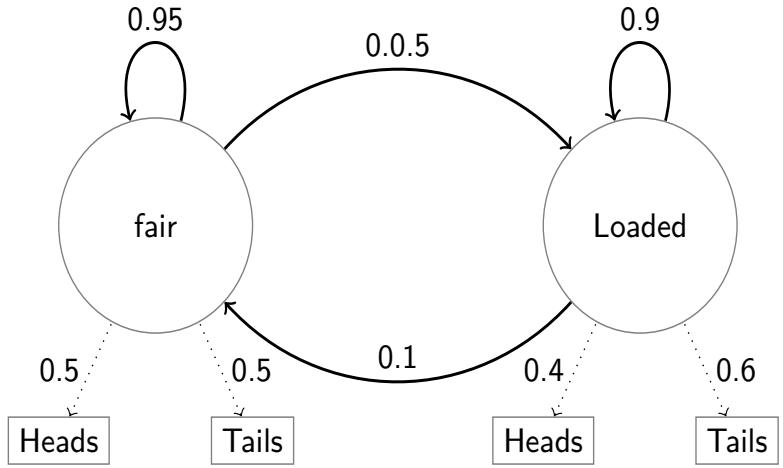
$$\mathbb{P}[\boldsymbol{Y}_k, \boldsymbol{X}_{k-1}|x_k = i] = \mathbb{P}[y_k|x_k = i]\mathbb{P}[x_k = i|x_{k-1}]\mathbb{P}[\boldsymbol{Y}_{k-1}, \boldsymbol{X}_{k-1}]$$

- **Recursion:**

$$\begin{aligned}
f(i,k) &= \max_{\boldsymbol{X}_{k-1}} \mathbb{P}[\boldsymbol{Y}_k, \boldsymbol{X}_{k-1}|x_k = i] \\
&= \max_{\boldsymbol{X}_{k-1}} \mathbb{P}[y_k|x_k = i]\mathbb{P}[x_k = i|x_{k-1}]\mathbb{P}[\boldsymbol{Y}_{k-1}, \boldsymbol{X}_{k-1}] \\
&= \mathbb{P}[y_k|x_k = i] \max_{\boldsymbol{X}_{k-1}} (\mathbb{P}[x_k = i|x_{k-1}]\mathbb{P}[\boldsymbol{Y}_{k-1}, \boldsymbol{X}_{k-1}]) \\
&= \mathbb{P}[y_k|x_k = i] \max_{j} \mathbb{P}[x_k = i|x_{k-1} = j] \max_{\boldsymbol{X}_{k-2}} \mathbb{P}[\boldsymbol{Y}_{k-1}, \boldsymbol{X}_{k-2}|x_{k-1} = j] \\
&= \mathbb{P}[y_k|x_k = i] \max_{j} A_{ij} f(j, k-1)
\end{aligned}$$

- For 2-state HMM, Viterbi's algorithm fills a $2 \times n$ table column by column with entries $f(i,k)$
- **Initialization:** Note that $f(i,1) = \mathbb{P}[y_1|x_1 = i]$ is known.

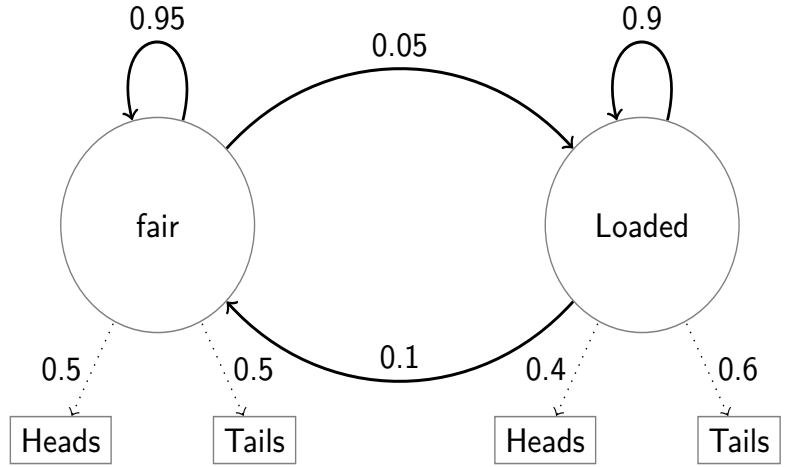
Viterbi's Algorithm: Example



- Find the most likely sequence of states given $\boldsymbol{Y}_6 = HTHTTT$.
- Instead of numbering states $i = 1, 2$, think $i = F, L$.

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$x_k = F$	0.5					
$x_k = L$	0.4					

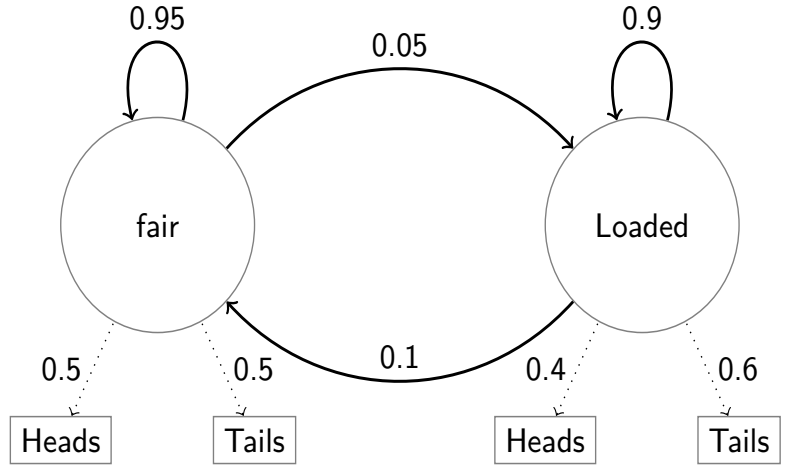
- $f(i,k) = \mathbb{P}[y_k | x_k = i] \max_j A_{ij} f(j, k-1)$
- $f(i,1) = \mathbb{P}[y_1 | x_1 = i]$



- $\boldsymbol{Y}_6 = HTHTTT$
- $f(i,k) = \mathbb{P}[y_k|x_k = i] \max_j A_{ij} f(j, k-1)$

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$x_k = F$	0.5	$\leftarrow 0.2375$				
$x_k = L$	0.4	$\leftarrow 0.216$				

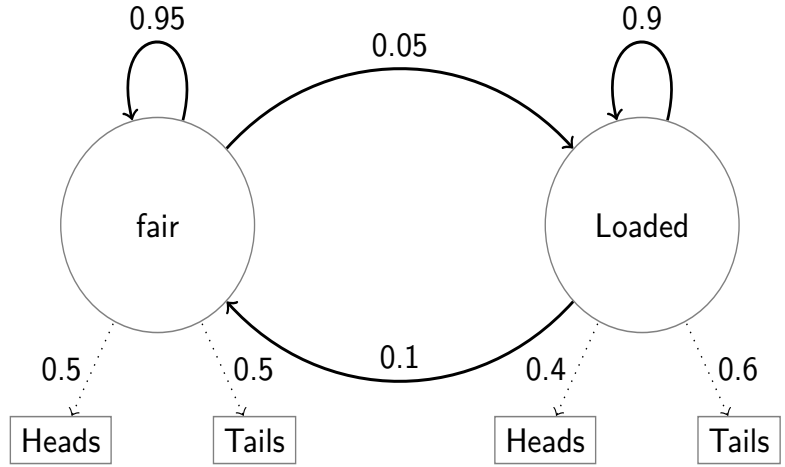
- $f(F,2) = \mathbb{P}[T|F] \max \begin{Bmatrix} 0.95 f(F,1) \\ 0.1 f(L,1) \end{Bmatrix} = (0.5) \max \begin{Bmatrix} \mathbf{0.475} \\ 0.04 \end{Bmatrix} = 0.2375$
- $f(L,2) = \mathbb{P}[T|L] \max \begin{Bmatrix} 0.05 f(F,1) \\ 0.9 f(L,1) \end{Bmatrix} = (0.6) \max \begin{Bmatrix} 0.025 \\ \mathbf{0.36} \end{Bmatrix} = 0.216$



- $\boldsymbol{Y}_6 = HTHTTT$
- $f(i,k) = \mathbb{P}[y_k|x_k = i] \max_j A_{ij} f(j, k-1)$

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$x_k = F$	0.5	$\leftarrow 0.2375$	$\leftarrow 0.1128$			
$x_k = L$	0.4	$\leftarrow 0.216$	$\leftarrow 0.0778$			

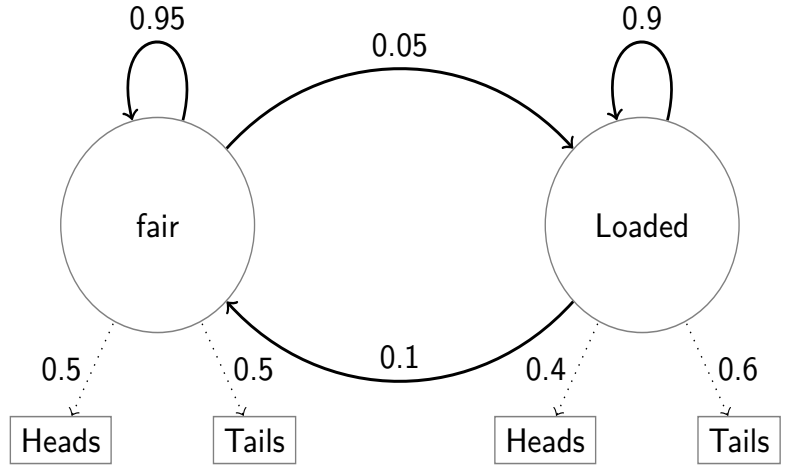
- $f(F,3) = \mathbb{P}[H|F] \max \begin{Bmatrix} 0.95 f(F,2) \\ 0.1 f(L,2) \end{Bmatrix} = (0.5) \max \begin{Bmatrix} \mathbf{0.2256} \\ 0.0216 \end{Bmatrix} = 0.1128$
- $f(L,3) = \mathbb{P}[H|L] \max \begin{Bmatrix} 0.05 f(F,2) \\ 0.9 f(L,2) \end{Bmatrix} = (0.4) \max \begin{Bmatrix} 0.0119 \\ \mathbf{0.1944} \end{Bmatrix} = 0.0778$



- $\boldsymbol{Y}_6 = HTHTTT$
- $f(i,k) = \mathbb{P}[y_k|x_k = i] \max_j A_{ij} f(j, k-1)$

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$x_k = F$	0.5	$\leftarrow 0.2375$	$\leftarrow 0.1128$	$\leftarrow 0.0536$		
$x_k = L$	0.4	$\leftarrow 0.216$	$\leftarrow 0.0778$	$\leftarrow 0.0420$		

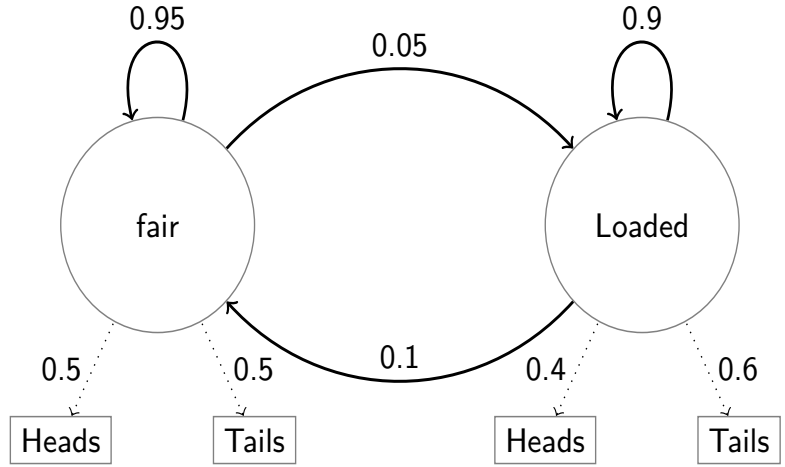
- $f(F,4) = \mathbb{P}[T|F] \max \begin{Bmatrix} 0.95 f(F,3) \\ 0.1 f(L,3) \end{Bmatrix} = (0.5) \max \begin{Bmatrix} \mathbf{0.1072} \\ 0.0078 \end{Bmatrix} = 0.0536$
- $f(L,4) = \mathbb{P}[T|L] \max \begin{Bmatrix} 0.05 f(F,3) \\ 0.9 f(L,3) \end{Bmatrix} = (0.6) \max \begin{Bmatrix} 0.0056 \\ \mathbf{0.0700} \end{Bmatrix} = 0.0420$



- $\boldsymbol{Y}_6 = HTHTTT$
- $f(i,k) = \mathbb{P}[y_k|x_k = i] \max_j A_{ij} f(j, k-1)$

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$x_k = F$	0.5	$\leftarrow 0.2375$	$\leftarrow 0.1128$	$\leftarrow 0.0536$	$\leftarrow 0.0255$	
$x_k = L$	0.4	$\leftarrow 0.216$	$\leftarrow 0.0778$	$\leftarrow 0.0420$	$\leftarrow 0.0227$	

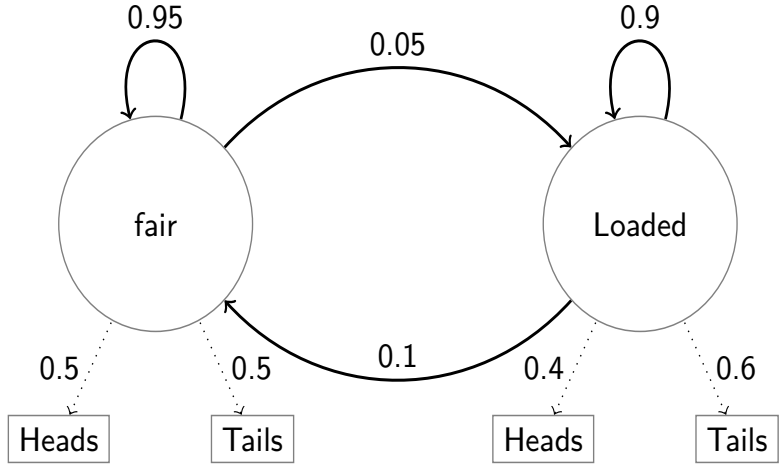
- $f(F,5) = \mathbb{P}[T|F] \max \begin{Bmatrix} 0.95 f(F,4) \\ 0.1 f(L,4) \end{Bmatrix} = (0.5) \max \begin{Bmatrix} \mathbf{0.0509} \\ 0.0042 \end{Bmatrix} = 0.0255$
- $f(L,5) = \mathbb{P}[T|L] \max \begin{Bmatrix} 0.05 f(F,4) \\ 0.9 f(L,4) \end{Bmatrix} = (0.6) \max \begin{Bmatrix} 0.0027 \\ \mathbf{0.0378} \end{Bmatrix} = 0.0227$



- $\boldsymbol{Y}_6 = HTHTTT$
- $f(i,k) = \mathbb{P}[y_k|x_k = i] \max_j A_{ij} f(j, k-1)$

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$x_k = F$	0.5	$\leftarrow 0.2375$	$\leftarrow 0.1128$	$\leftarrow 0.0536$	$\leftarrow 0.0255$	$\leftarrow 0.0121$
$x_k = L$	0.4	$\leftarrow 0.216$	$\leftarrow 0.0778$	$\leftarrow 0.0420$	$\leftarrow 0.0227$	$\leftarrow \mathbf{0.0123}$

- $f(F,6) = \mathbb{P}[T|F] \max \begin{Bmatrix} 0.95 f(F,5) \\ 0.1 f(L,5) \end{Bmatrix} = (0.5) \max \begin{Bmatrix} \mathbf{0.0242} \\ 0.0022 \end{Bmatrix} = 0.0121$
- $f(L,6) = \mathbb{P}[T|L] \max \begin{Bmatrix} 0.05 f(F,5) \\ 0.9 f(L,5) \end{Bmatrix} = (0.6) \max \begin{Bmatrix} 0.0013 \\ \mathbf{0.0204} \end{Bmatrix} = 0.0123$



- $\boldsymbol{Y}_6 = HTHTTT$
- $f(i,k) = \mathbb{P}[y_k|x_k = i] \max_j A_{ij} f(j, k-1)$

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$x_k = F$	0.5	$\leftarrow$ 0.2375	$\leftarrow$ 0.1128	$\leftarrow$ 0.0536	$\leftarrow$ 0.0255	$\leftarrow$ 0.0121
$x_k = L$	0.4	$\leftarrow$ 0.216	$\leftarrow$ 0.0778	$\leftarrow$ 0.0420	$\leftarrow$ 0.0227	$\leftarrow$ **0.0123**

- Note that the final answer is the max entry in final column, **not necessarily** the bottom right hand corner.
- Traceback: The most likely sequence of states is LLLLLL.

Summary: Dynamic Programming

- Dynamic Programming is an algorithmic technique for discrete optimization.
- Works for problems that can be divided into stages, with a policy decision/ choice (*e.g. "add item to bag" or "decide if dice is fair or loaded"* to be made at each stage.
- Need to define a **value function** $f(i, k)$, interpreted as "best possible outcome up to k-th stage, given that we make decision i at k-th stage.
- The optimal choices for stages $k + 1, \ldots, n$ depend only on the choice made at stage k, and not on previous policy decisions.
- **Hardest Part:** Finding a recursive definition for $f(i, k)$.
- **Traceback** is a method for finding the optimal policy given the value of the optimal policy $\max_i f(i, k)$